

```
import pandas as pd

df = pd.read_csv("Dataset .csv")

print(df.shape)
df.head()
```

... (9551, 21)

Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Ta hook
------------------	--------------------	-----------------	------	---------	----------	---------------------	-----------	----------	----------	-----	----------	------------

```
import pandas as pd

df = pd.read_csv("Dataset .csv")

print(df.shape)
df.head()
```

... (9551, 21)

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)	Yes	No	No	No	\$
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)	Yes	No	No	No	\$
2	6300002	Heat - Edsa	162	Mandaluyong	Edsa Shangri-La, 1 Garden	Edsa Shangri-La, Ortigas	Edsa Shangri-La, Ortigas	121.056834	14.581404	Seafood, Asian,		Botswana	Yes	No	No	No	\$



df.info()

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Restaurant ID                        9551 non-null   int64
1   Restaurant Name                      9551 non-null   object
2   Country Code                        9551 non-null   int64
3   City                                9551 non-null   object
4   Address                             9551 non-null   object
5   Locality                            9551 non-null   object
6   Locality Verbose                    9551 non-null   object
7   Longitude                           9551 non-null   float64
8   Latitude                            9551 non-null   float64
9   Cuisines                            9542 non-null   object
10  Average Cost for two                 9551 non-null   int64
11  Currency                            9551 non-null   object
12  Has Table booking                   9551 non-null   object
13  Has Online delivery                 9551 non-null   object
14  Is delivering now                   9551 non-null   object
15  Switch to order menu                9551 non-null   object
16  Price range                         9551 non-null   int64
17  Aggregate rating                    9551 non-null   float64
18  Rating color                        9551 non-null   object
19  Rating text                         9551 non-null   object
20  Votes                              9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```



```
df.isnull().sum()
```

...

0

Restaurant ID

0

Restaurant Name

0

Country Code

0

City

0

Address

0

Locality

0

Locality Verbose

0

Longitude

0

Latitude

0

Cuisines

9

Average Cost for two

0



```
df.isnull().sum()
```

...

0

Restaurant ID

0

Restaurant Name

0

Country Code

0

City

0

Address

0

Locality

0

Locality Verbose

0

Longitude

0

Latitude

0

Cuisines

9

Average Cost for two

0

Currency

0

Has Table booking

0

Has Online delivery

0

Is delivering now

0

Switch to order menu 0

```
df["Cuisines"] = df["Cuisines"].fillna("Unknown")
```

```
df.isnull().sum()
```

0

Restaurant ID	0
----------------------	---

Restaurant Name	0
------------------------	---

Country Code	0
---------------------	---

City	0
-------------	---

Address	0
----------------	---

Locality	0
-----------------	---

Locality Verbose	0
-------------------------	---

Longitude	0
------------------	---

Latitude	0
-----------------	---

Cuisines	0
-----------------	---

Average Cost for two	0
-----------------------------	---

Currency	0
-----------------	---

Has Table booking	0
--------------------------	---

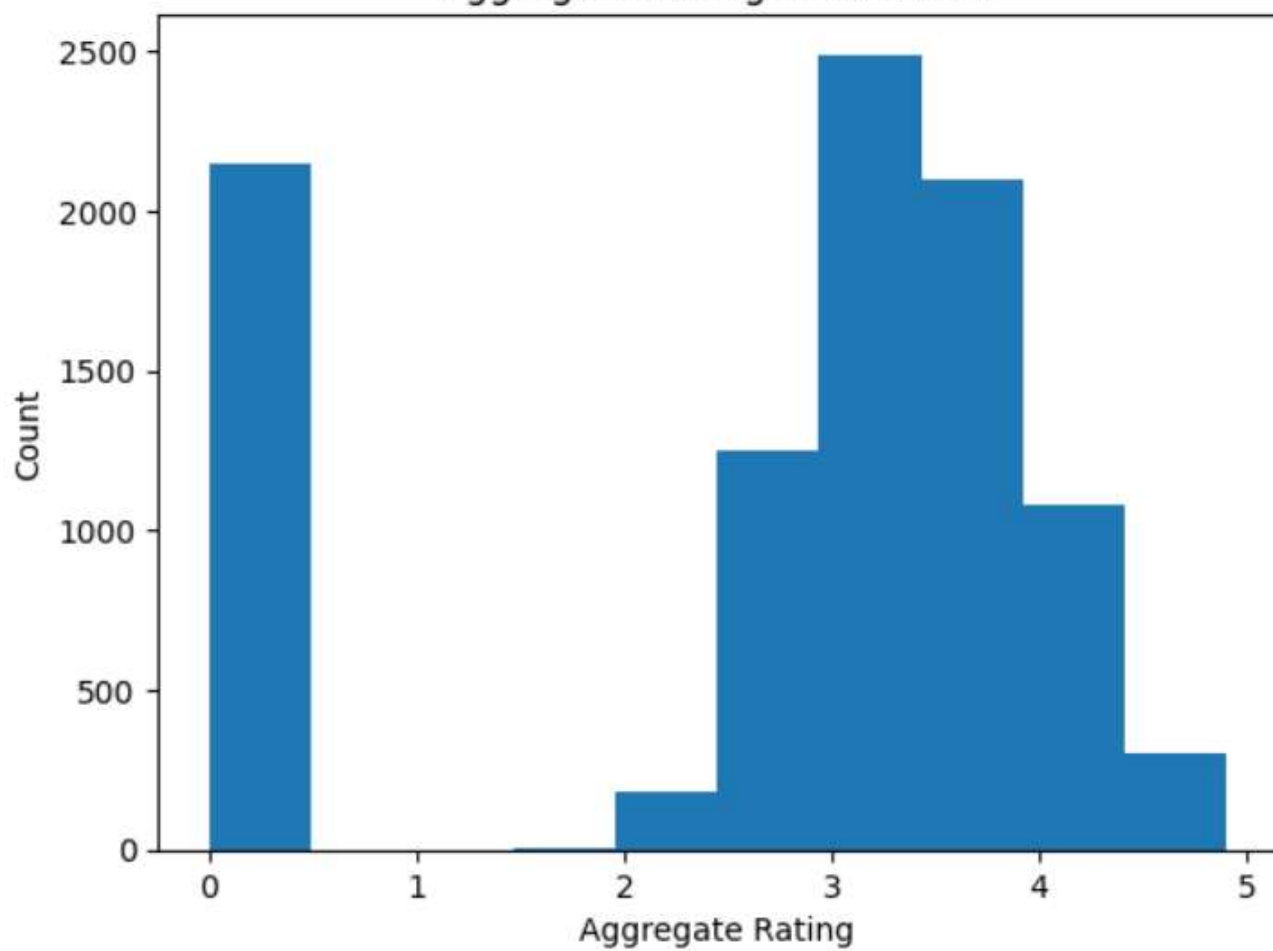
Has Online delivery	0
----------------------------	---



```
import matplotlib.pyplot as plt

plt.hist(df["Aggregate rating"])
plt.title("Aggregate Rating Distribution")
plt.xlabel("Aggregate Rating")
plt.ylabel("Count")
plt.show()
```

Aggregate Rating Distribution



 `import pandas as pd`

`df = pd.read_csv("Dataset .csv")`

`df.describe()`

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.666370	156.909748
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.516378	430.169145
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.500000	5.000000
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.200000	31.000000
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.700000	131.000000
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.900000	10934.000000

100% | 100% | 100% | 100% | 100%



```
df["Country Code"].value_counts()
```

..

count

Country Code

1	8652
---	------

216	434
-----	-----

215	80
-----	----

30	60
----	----

189	60
-----	----

214	60
-----	----

148	40
-----	----

208	34
-----	----

14	24
----	----

162	22
-----	----

94	21
----	----

166	20
-----	----

184	20
-----	----

191	20
-----	----

37	4
----	---

```
df["Cuisines"].value_counts().head(10)
```

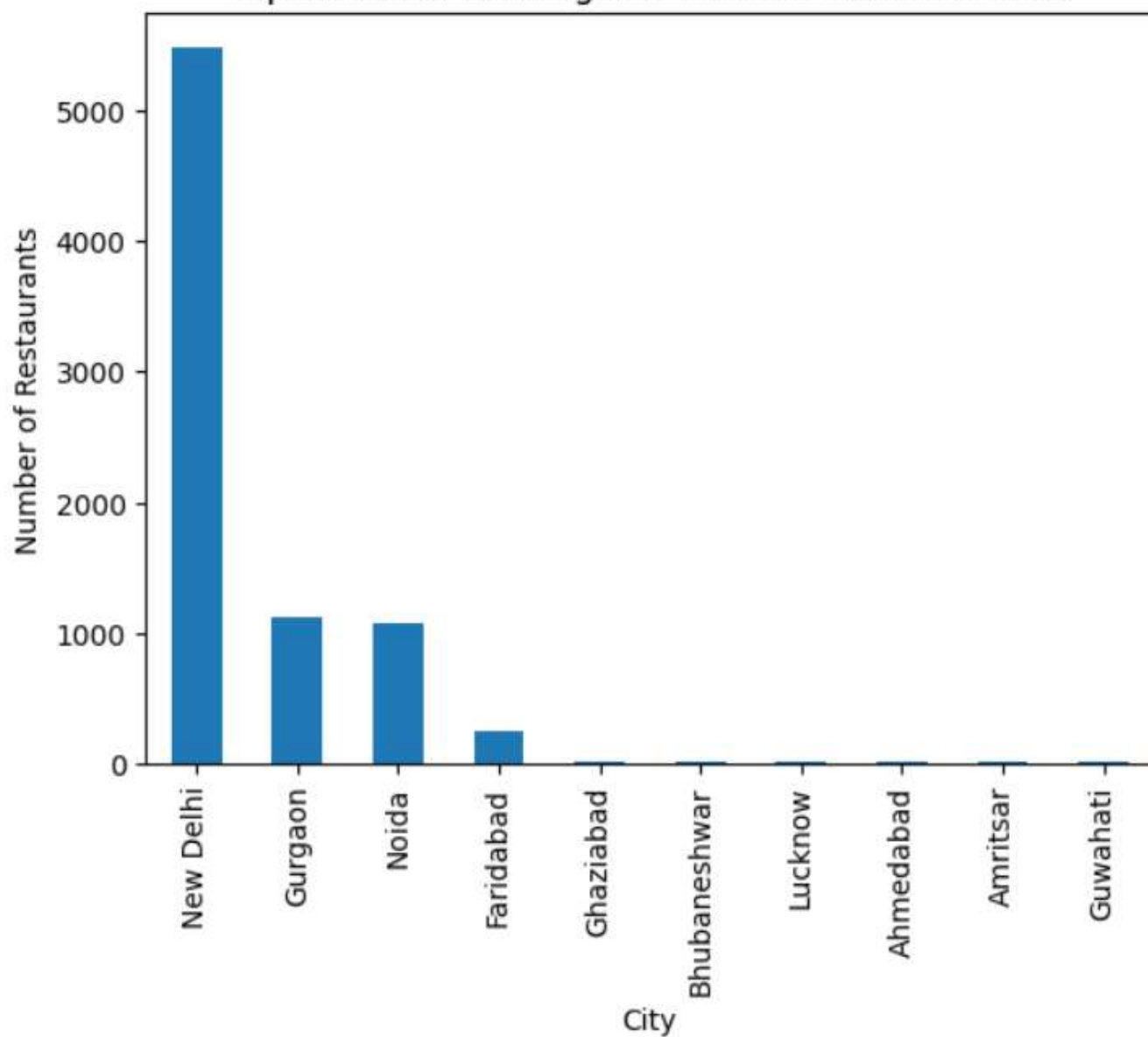
	count
Cuisines	
North Indian	936
North Indian, Chinese	511
Chinese	354
Fast Food	354
North Indian, Mughlai	334
Cafe	299
Bakery	218
North Indian, Mughlai, Chinese	197
Bakery, Desserts	170
Street Food	149

dtype: int64

▶ `import matplotlib.pyplot as plt`

```
df["City"].value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Cities with Highest Number of Restaurants")
plt.xlabel("City")
plt.ylabel("Number of Restaurants")
plt.show()
```

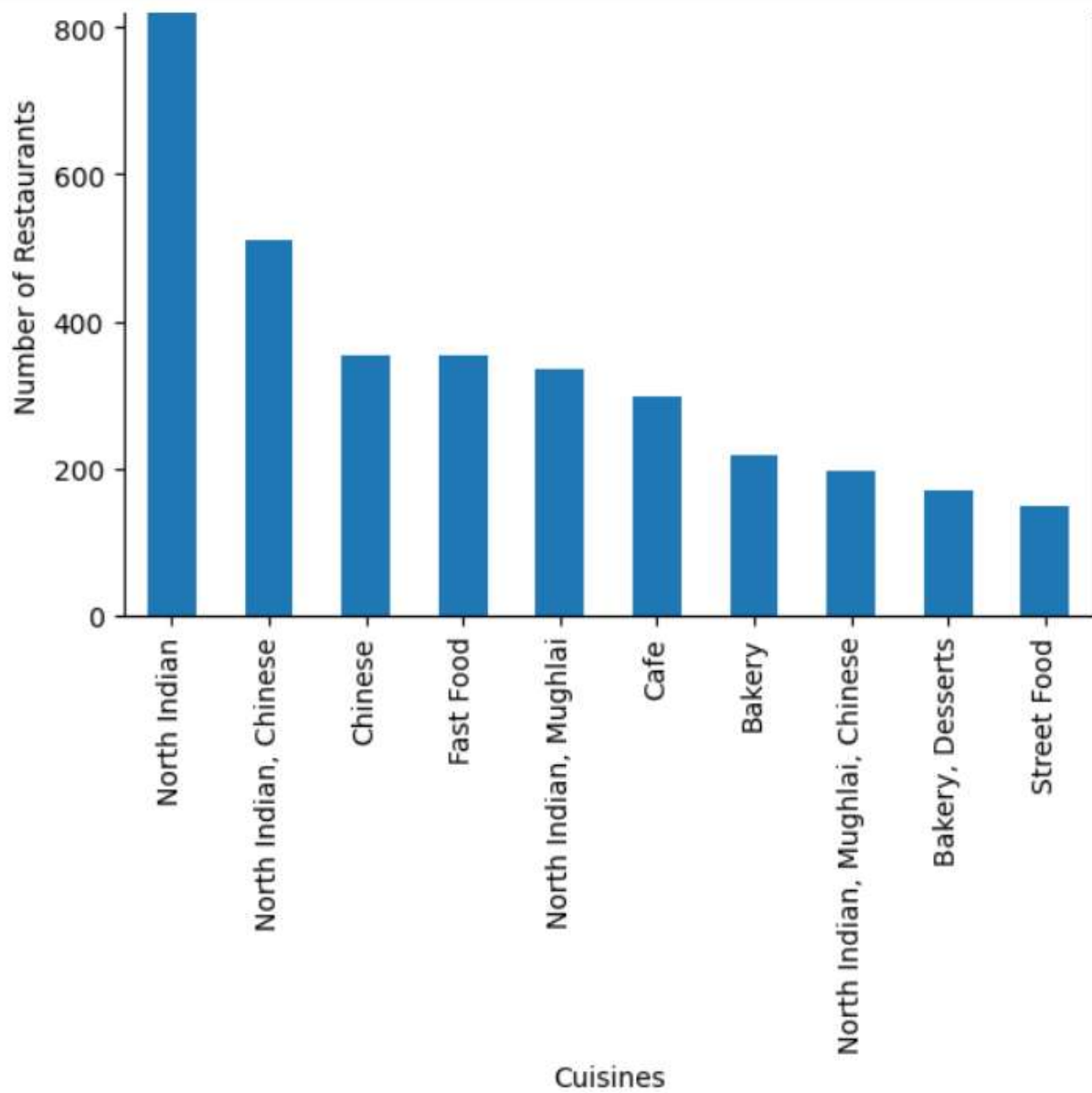
Top 10 Cities with Highest Number of Restaurants



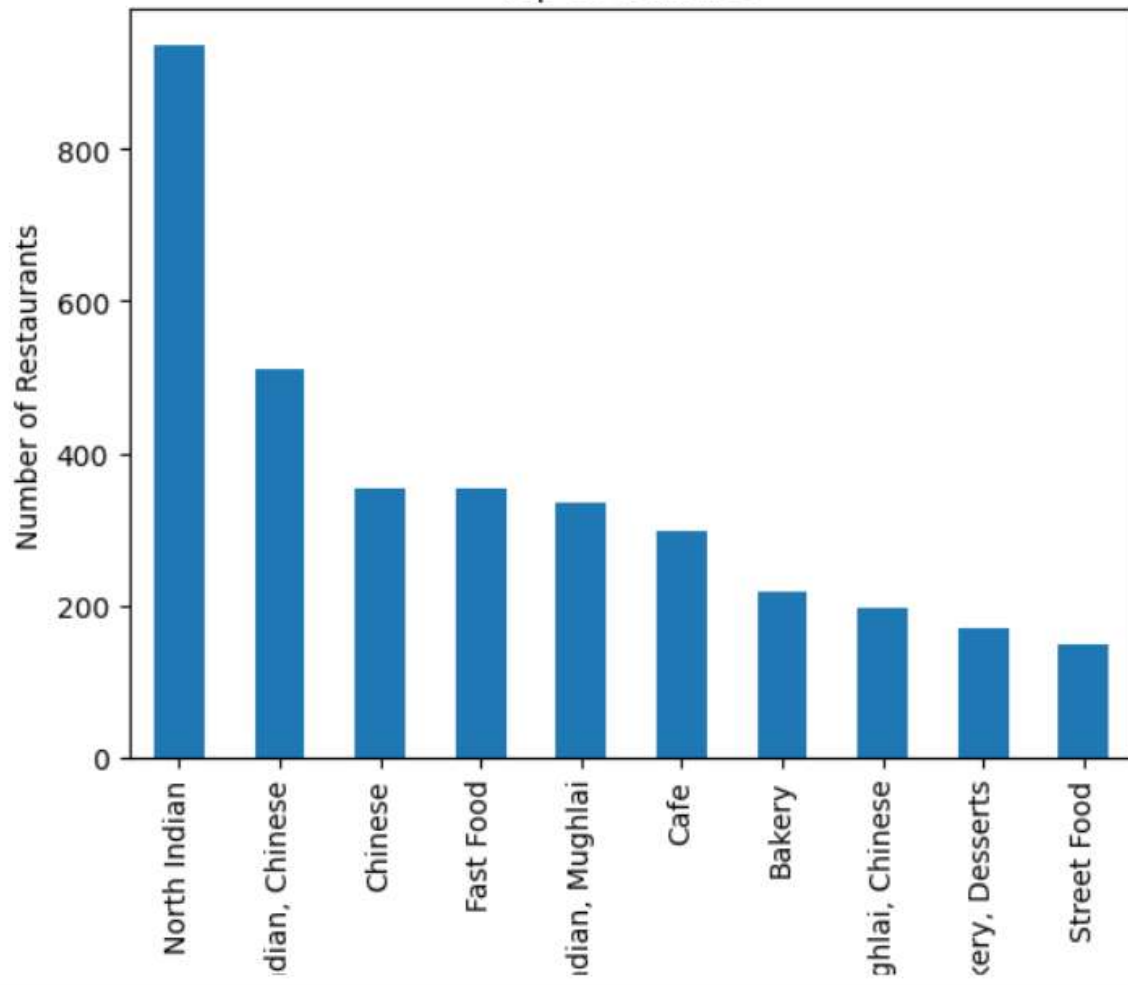
```
import matplotlib.pyplot as plt
```

```
df["Cuisines"].value_counts().head(10).plot(kind="bar")  
plt.title("Top 10 Cuisines")  
plt.xlabel("Cuisines")  
plt.ylabel("Number of Restaurants")  
plt.show()
```

..



Top 10 Cuisines




```
import pandas as pd
import folium
```

```
df = pd.read_csv("Dataset .csv")
```

```
▶ m = folium.Map(location=[df["Latitude"].mean(),df["Longitude"].mean()],zoom_start=5)
for i in range(100):
    folium.Marker([df.iloc[i]["Latitude"],df.iloc[i]["Longitude"]]).add_to(m)

display(m)
```







```
country_count = df[ "Country Code" ].value_counts()  
print(country_count)
```

```
city_count = df[ "City" ].value_counts()  
print(city_count.head(10))
```

... Country Code

1	8652
216	434
215	80
30	60
189	60
214	60
148	40
208	34
14	24
162	22
94	21
166	20
184	20
191	20
37	4

Name: count, dtype: int64

City

New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Bhubaneswar	21
Lucknow	21
Ahmedabad	21
Amritsar	21
Guwahati	21

Name: count, dtype: int64

```
correlation = df[["Latitude", "Longitude", "Aggregate rating"]].corr()  
print(correlation)
```

```
...  
Latitude      Latitude  Longitude  Aggregate rating  
Longitude      0.043207   1.000000      -0.116818  
Aggregate rating 0.000516  -0.116818      1.000000
```

```
import pandas as pd
df = pd.read_csv("Dataset .csv")
table_booking_percentage = (df["Has Table booking"].value_counts(normalize=True) * 100)

print(table_booking_percentage)
```

```
... Has Table booking
No    87.875615
Yes   12.124385
Name: proportion, dtype: float64
```



```
online_delivery_percentage = (df["Has Online delivery"].value_counts(normalize=True) * 100)

print(online_delivery_percentage)
```

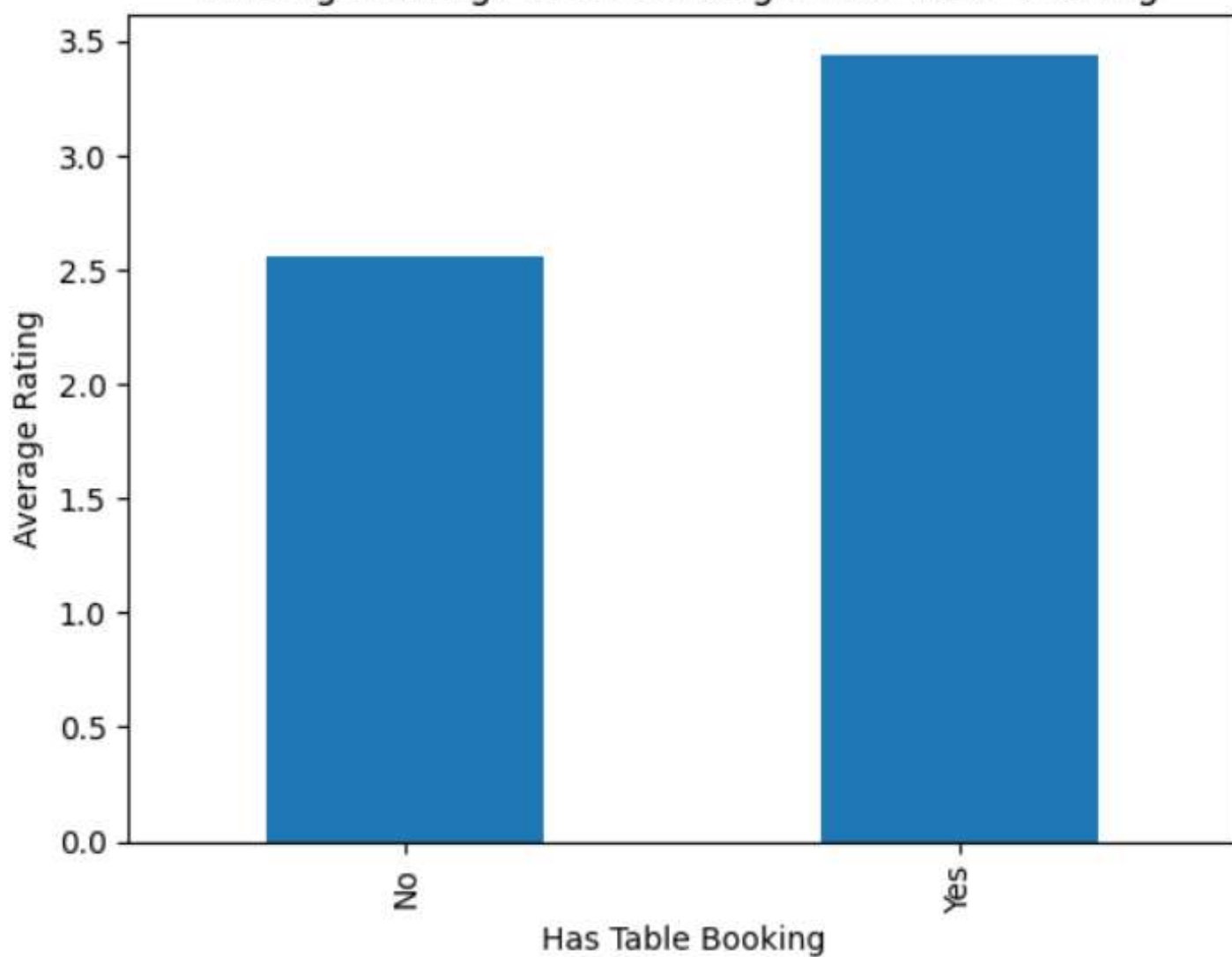
```
Has Online delivery
No      74.337766
Yes     25.662234
Name: proportion, dtype: float64
```

```
average_rating_table_booking = df.groupby("Has Table booking")["Aggregate rating"].mean()

print(average_rating_table_booking)
```

```
... Has Table booking
No    2.559359
Yes    3.441969
Name: Aggregate rating, dtype: float64
```

Average Rating: Table Booking vs No Table Booking



```
▶ online_delivery_price = pd.crosstab(  
    df["Price range"],  
    df["Has Online delivery"],  
    normalize="index"  
) * 100  
  
print(online_delivery_price)
```

...	Has Online delivery	No	Yes
	Price range		
	1	84.225923	15.774077
	2	58.689367	41.310633
	3	70.809659	29.190341
	4	90.955631	9.044369

```
import matplotlib.pyplot as plt

online_delivery_price = pd.crosstab(
    df["Price range"],
    df["Has Online delivery"],
    normalize="index"
) * 100

online_delivery_price.plot(kind="bar")

plt.title("Online Delivery Availability by Price Range")
plt.xlabel("Price Range")
plt.ylabel("Percentage of Restaurants")
plt.legend(title="Online Delivery")
plt.show()
```

Online Delivery Availability by Price Range



```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("Dataset .csv")
```

```
▶ price_count = df["Price range"].value_counts()

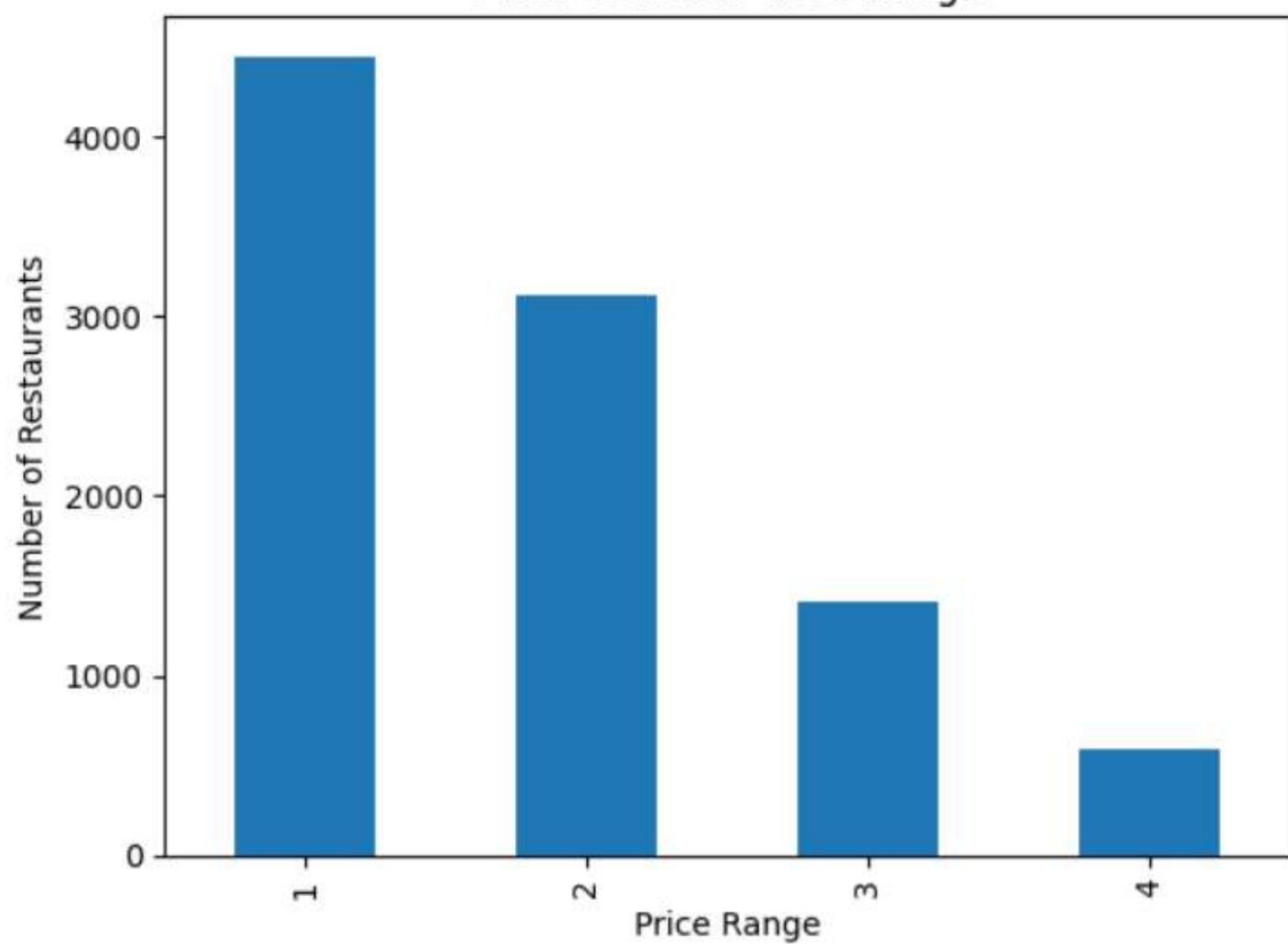
print(price_count)
```

```
... Price range
1      4444
2      3113
3      1408
4       586
Name: count, dtype: int64
```

```
price_count.plot(kind="bar")
plt.title("Most Common Price Range")
plt.xlabel("Price Range")
plt.ylabel("Number of Restaurants")
plt.show()
```

...

Most Common Price Range

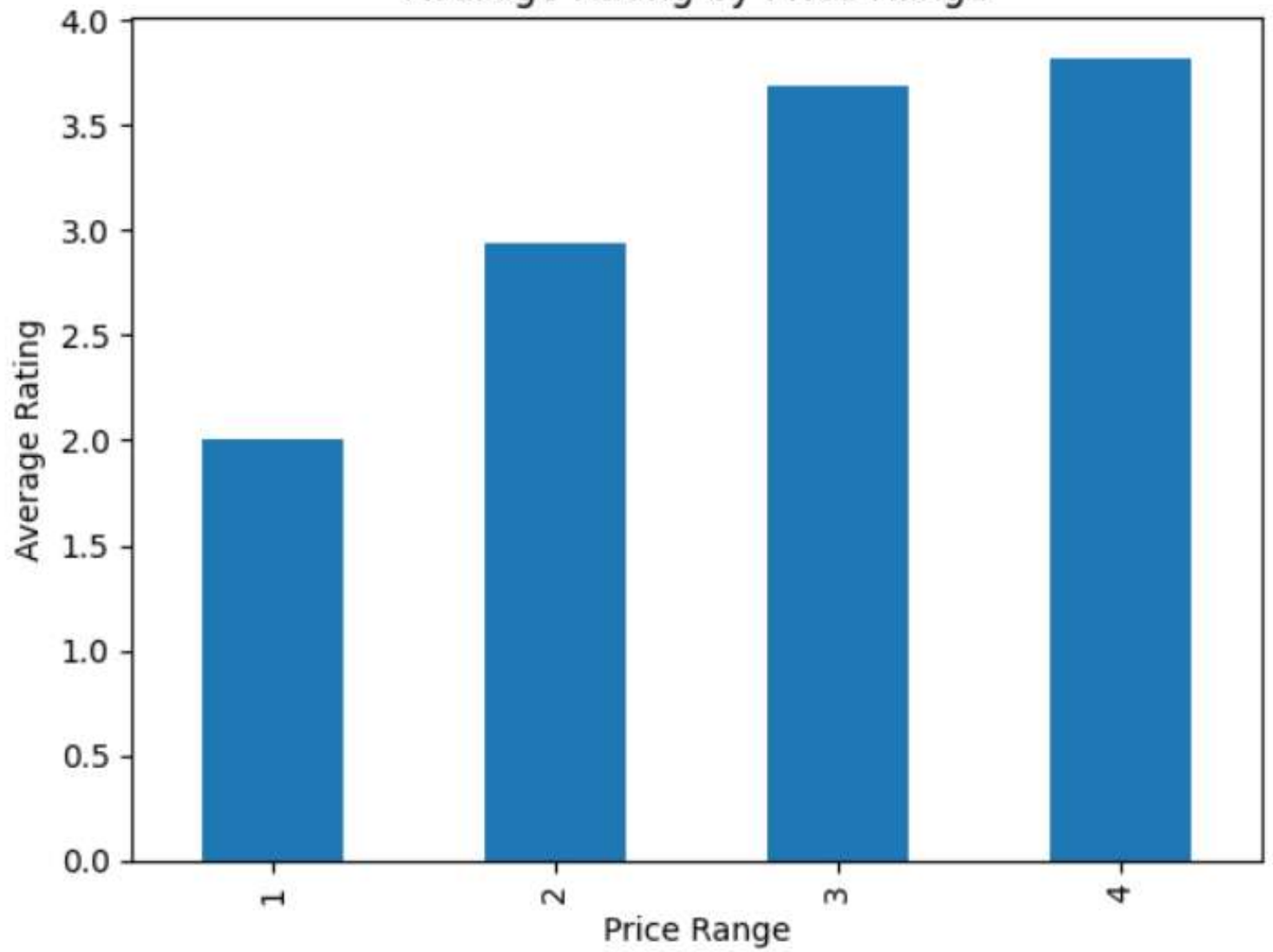



```
▶ average_rating = df.groupby("Price range")["Aggregate rating"].mean()  
  
print(average_rating)
```

```
... Price range  
1      1.999887  
2      2.941054  
3      3.683381  
4      3.817918  
Name: Aggregate rating, dtype: float64
```

```
average_rating.plot(kind="bar")  
  
plt.title("Average Rating by Price Range")  
plt.xlabel("Price Range")  
plt.ylabel("Average Rating")  
  
plt.show()
```

Average Rating by Price Range

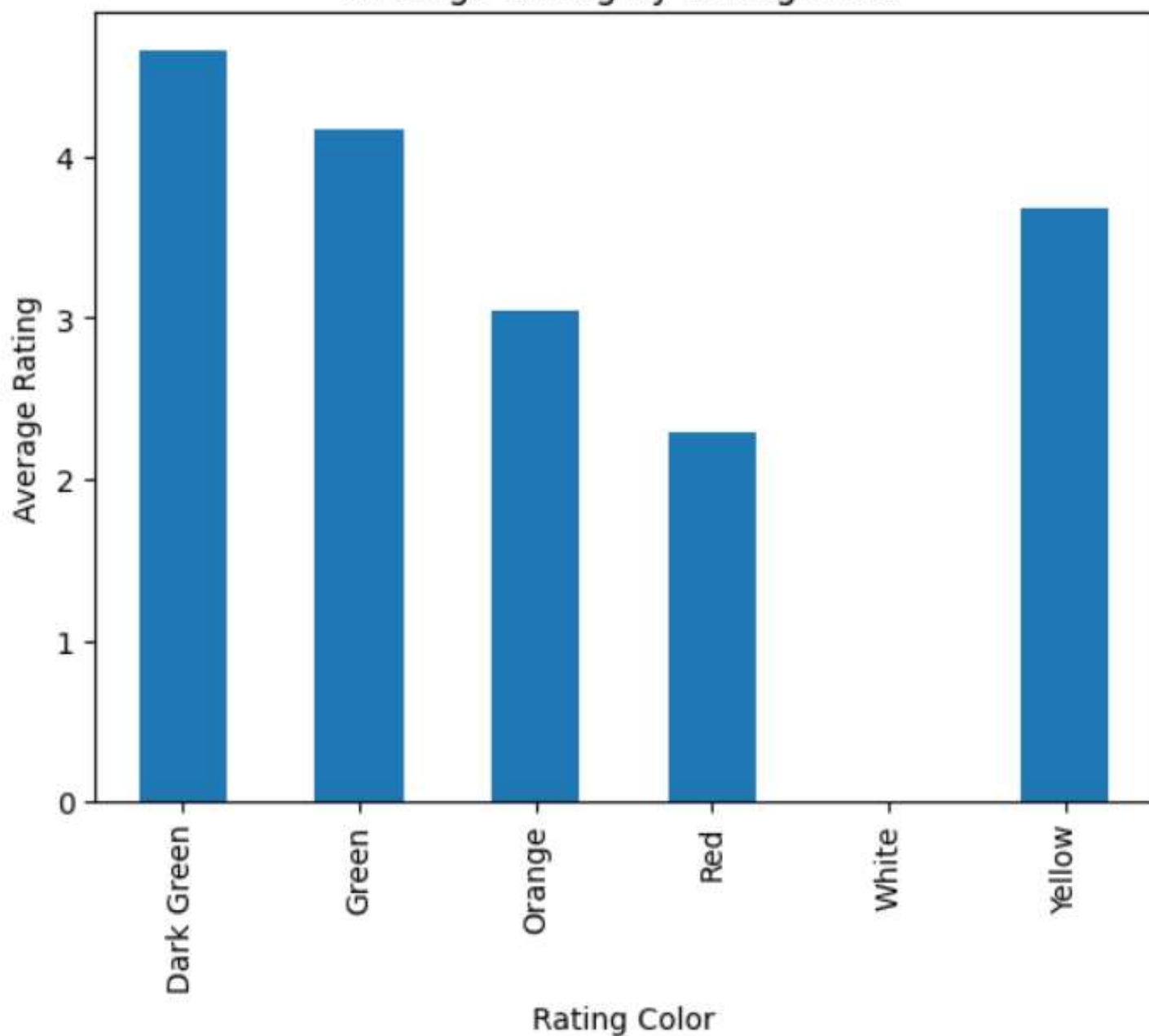


```
rating_color = df.groupby("Rating color")["Aggregate rating"].mean()  
  
print(rating_color)
```

```
Rating color  
Dark Green    4.659801  
Green         4.168119  
Orange        3.051619  
Red           2.297849  
White         0.000000  
Yellow        3.683429  
Name: Aggregate rating, dtype: float64
```

```
▶ rating_color.plot(kind="bar")  
  
plt.title("Average Rating by Rating Color")  
plt.xlabel("Rating Color")  
plt.ylabel("Average Rating")  
  
plt.show()
```

Average Rating by Rating Color



```
import pandas as pd
```

```
df = pd.read_csv("Dataset .csv")
```

```
df["Restaurant Name Length"] = df["Restaurant Name"].str.len()
df["Address Length"] = df["Address"].str.len()

print(df[["Restaurant Name", "Restaurant Name Length",
          "Address", "Address Length"]].head())
```

```
...      Restaurant Name  Restaurant Name Length \
0      Le Petit Souffle                      16
1      Izakaya Kikufuji                      16
2  Heat - Edsa Shangri-La                    22
3                      Ooma                     4
4      Sambo Kojin                          11

                                Address  Address Length
0  Third Floor, Century City Mall, Kalayaan Avenu...      71
1  Little Tokyo, 2277 Chino Roces Avenue, Legaspi...      67
2  Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...      56
3  Third Floor, Mega Fashion Hall, SM Megamall, O...      70
4  Third Floor, Mega Atrium, SM Megamall, Ortigas...
```

▶ `import matplotlib.pyplot as plt`

`sample = df.head(10)`

`plt.figure(figsize=(10,5))`

`plt.bar(sample["Restaurant Name"], sample["Restaurant Name Length"])`

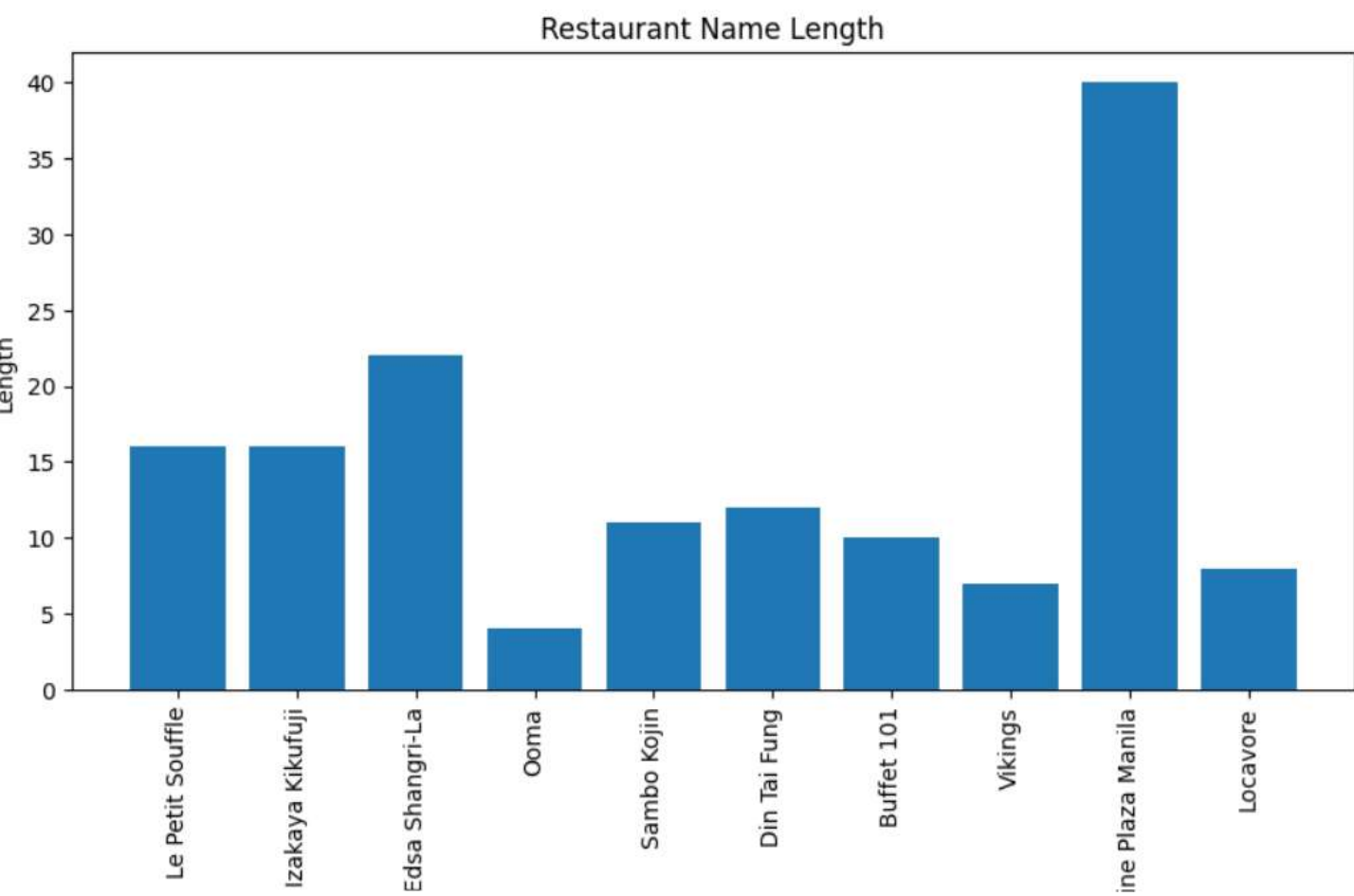
`plt.xticks(rotation=90)`

`plt.title("Restaurant Name Length")`

`plt.xlabel("Restaurant Name")`

`plt.ylabel("Length")`

`plt.show()`



```
df["Has Table Booking"] = df["Has Table booking"].map({"Yes": 1, "No": 0})
df["Has Online Delivery"] = df["Has Online delivery"].map({"Yes": 1, "No": 0})

print(df[["Has Table booking", "Has Table Booking",
          "Has Online delivery", "Has Online Delivery"]].head())
```

	Has Table booking	Has Table Booking	Has Online delivery	\
0	Yes	1	No	
1	Yes	1	No	
2	Yes	1	No	
3	No	0	No	
4	Yes	1	No	

	Has Online Delivery
0	0
1	0
2	0
3	0
4	0

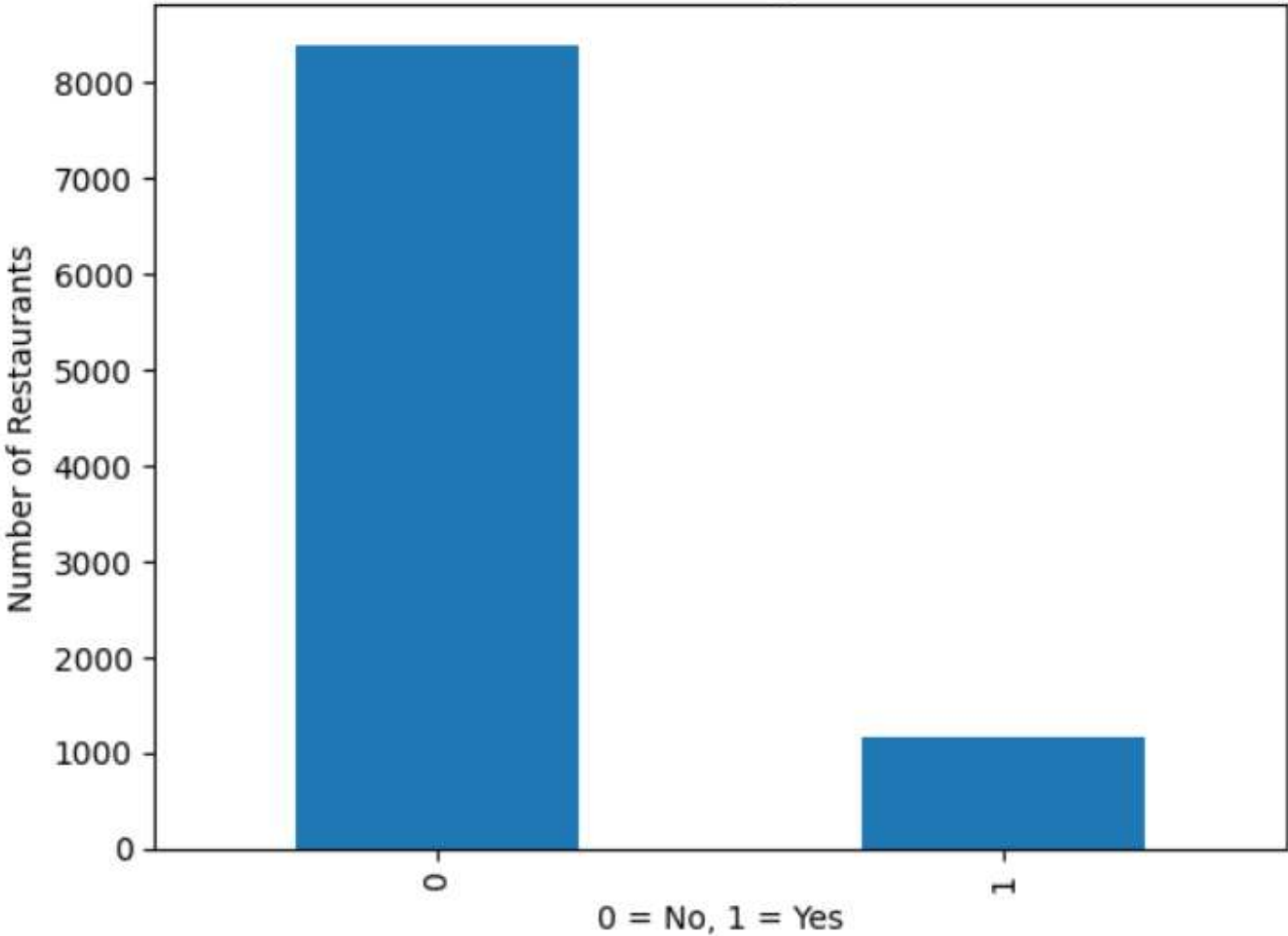
```
import matplotlib.pyplot as plt
```

```
booking_count = df["Has Table Booking"].value_counts()
```

```
booking_count.plot(kind="bar")
plt.title("Has Table Booking (Encoded)")
plt.xlabel("0 = No, 1 = Yes")
plt.ylabel("Number of Restaurants")
plt.show()
```


...

Has Table Booking (Encoded)



```
▶ delivery_count = df["Has Online Delivery"].value_counts()

delivery_count.plot(kind="bar")
plt.title("Has Online Delivery (Encoded)")
plt.xlabel("0 = No, 1 = Yes")
plt.ylabel("Number of Restaurants")
plt.show()
```

...

...

Has Online Delivery (Encoded)

